Running the IMERG workflow as a Python script

User guide for **imerg_basin_precip.py** — download IMERG GPM Final Run data and compute basin-averaged precipitation without Jupyter

1. What the script does

`imerg_basin_precip.py` is the operational version of `IMERG_notebook.ipynb`. It runs the core steps of the notebook in one go, without widgets or interactive maps:

- checks the IMERG files already available in the **nc/** folder and, only if some timesteps are missing, reads your NASA Earthdata credentials from **.netrc** and downloads them;
- crops the IMERG grid to the area of interest and converts precipitation to **depth per timestep (mm)**, exactly as in the notebook;
- loads the basin shapefile from **shapefile/** and computes the basin-averaged precipitation with the three methods (area-weighted, all_touched=True, all_touched=False);
- lets you choose the method, then saves the time series to an **Excel file** and a **PNG plot** in **output/**.

It works with all three IMERG temporal resolutions: **30-minute** (GPM_3IMERGHH), **daily** (GPM_3IMERGDF) and **monthly** (GPM_3IMERGM), version 07.

2. Before you start

The script uses the same setup as the notebook. If you have already followed `IMERG_setup.pdf`, you are ready. Otherwise, follow that guide to:

- create the conda environment **imerg_env** from `environment.yml` (no extra libraries are needed for the script);
- create a NASA Earthdata account and authorize the **NASA GES DISC** application in your profile;
- create the `.netrc` file in the repository folder (the same folder as the notebook and the script).

Folder structure

All paths are relative to the folder that contains the script, so the script can be launched from any directory. The folders are the same used by the notebook:

```
repository_folder/
├── imerg_basin_precip.py
├── IMERG_notebook.ipynb
├── .netrc              <- NASA Earthdata credentials
├── shapefile/          <- exactly ONE basin shapefile (.shp, .shx, .dbf, .prj, ...)
├── nc/                 <- IMERG files (created automatically)
│   ├── 30_min/
│   ├── Daily/
│   └── Monthly/
└── output/             <- Excel files and plots (created automatically)
```

> Files downloaded by the notebook and by the script are stored in the same `nc/` subfolders, so data downloaded once can be reused by both.

3. Running the script step by step

Step 1 — Open a terminal. On Windows, open **Anaconda Prompt** from the Start menu. On macOS or Linux, open the **Terminal**.

Step 2 — Activate the course environment imerg_env.

```
conda activate imerg_env
```

The beginning of the command line changes from **(base)** to **(imerg_env)**.

> **Always run the script inside imerg_env**, the same environment used for the notebook. The **(base)** environment does not contain the required libraries (xarray, geopandas, earthaccess, ...) and the script would stop with a *ModuleNotFoundError*. Repeat this step every time you open a new terminal.

Step 3 — Move to the repository folder (adapt the path to where you saved the repository):

```
cd Desktop\imerg_course          (Windows)
cd ~/Desktop/imerg_course         (macOS / Linux)
```

Step 4 — Run the script with the default settings:

```
python imerg_basin_precip.py
```

Step 5 — Choose the basin-averaging method. The script prints a summary of the three methods (number of IMERG cells used and total precipitation over the period) and asks which one to keep. Type 1, 2 or 3, or press **Enter** for the area-weighted method:

```
Basin-averaging methods
  #  method                cells   total (mm)
  1  Area-weighted            ..         ....
  2  all_touched=True         ..         ....
  3  all_touched=False        ..         ....

Choose the method [1-3, Enter = Area-weighted]:
```

A method that selects no IMERG cell (for example all_touched=False on a very small basin) is shown as *n/a* and cannot be chosen.

Step 6 — Open the results in the **output/** folder (see Section 6).

4. Changing the settings

There are two equivalent ways to change the analysis settings.

Option A — edit the SETTINGS block

Open the script with any text editor (Notepad, VS Code, Jupyter Lab, ...) and edit the block at the top of the file, then save and run the script again:

```
TIME_SCALE = "Daily"          # "30 min", "Daily" or "Monthly"
START_DATE = "2023-06-01"     # YYYY-MM-DD, included
END_DATE   = "2023-06-30"     # YYYY-MM-DD, included
BBOX = None                   # (lon_min, lat_min, lon_max, lat_max) or None
BBOX_BUFFER = 0.2             # degrees added around the basin when BBOX = None
METHOD = None                 # "Area-weighted", "all_touched=True",
                              # "all_touched=False" or None (= ask)
SHOW_PLOT = False             # True = also show the plot on screen
```

Option B — command-line options

Options written after the script name override the SETTINGS block for that run only:

Option	Meaning	Accepted values
`--scale`	Temporal resolution	30min, Daily, Monthly
`--start`	Start date (included)	YYYY-MM-DD
`--end`	End date (included)	YYYY-MM-DD
`--bbox`	Area of interest (four numbers)	LON_MIN LAT_MIN LON_MAX LAT_MAX
`--method`	Basin-averaging method (skips the question)	weighted, true, false
`--show`	Also show the plot on screen	(no value)
`--help`	Print the list of options	(no value)

Examples

```
# Monthly precipitation 2020-2023, method chosen interactively
python imerg_basin_precip.py --scale Monthly --start 2020-01-01 --end 2023-12-31

# 30-minute precipitation during an event, area-weighted method
python imerg_basin_precip.py --scale 30min --start 2023-06-10 --end 2023-06-11 --method weighted

# Daily precipitation over a user-defined area, plot shown on screen
python imerg_basin_precip.py --scale Daily --bbox 12.0 42.5 13.0 43.5 --show
```

5. The area of interest

If `BBOX` is **None** (default), the area is computed automatically from the shapefile extent plus `BBOX_BUFFER` degrees on each side (0.2° = two IMERG cells). For the La_Bruna example catchment you do not need to define anything.

If you set `BBOX` or use `--bbox`, the area must contain the whole basin: otherwise the script stops with a message asking you to enlarge it. Coordinates are in decimal degrees (WGS84), with longitude first.

6. Outputs

Each run writes two files in **output/**. The file name contains the temporal resolution, the method and the period, so different runs do not overwrite each other:

```
basin_precipitation_Daily_Area-weighted_20230601_20230630.xlsx
basin_precipitation_Daily_Area-weighted_20230601_20230630.png
```

File	Content
Excel — sheet **precipitation**	`date` and `basin_averaged_precipitation_mm`: one row per timestep, precipitation depth in mm per timestep (30 minutes, 1 day or 1 month). Missing values are left empty, never replaced by zero.
Excel — sheet **info**	Settings and summary of the run: product and version, period, area, basin area (km²), method, number of IMERG cells used, missing timesteps, total precipitation.
PNG plot	Basin-averaged precipitation time series, saved at 300 dpi.

7. Good to know

- **Download only when needed.** The Earthdata login happens only if some timesteps are missing locally. If all files are already in nc/, the script also works offline.
- **Data volume.** Every IMERG file covers the whole globe, even when you select a small area. One month of 30-minute data means about 1440 files: start with short periods.
- **Units.** 30-minute data (mm/hr) are multiplied by 0.5; daily data are already in mm/day; monthly data (mm/hr) are multiplied by 24 and by the number of days in the month.
- **Dates.** The time of each value is read from the IMERG file name (start of the interval, UTC).

8. Troubleshooting

Message / problem	What to do
ModuleNotFoundError: No module named 'xarray' (or another library)	The script was launched outside the course environment: run `conda activate imerg_env` and try again (the prompt must start with (imerg_env)).
.netrc file not found	Create `.netrc` in the same folder as the script (see IMERG_setup.pdf).
no 'machine urs.earthdata.gov' entry	The first line of `.netrc` must be `machine urs.earthdata.gov`.
EulaNotAccepted during download	Log in to your Earthdata profile and authorize the NASA GES DISC application (see IMERG_setup.pdf, Step 4).
found N .shp files in 'shapefile/'	Keep exactly one basin shapefile in the shapefile/ folder.
the shapefile contains N features	The basin must be a single polygon: merge the features with a dissolve in GIS software.
the IMERG grid does not fully cover the basin	Enlarge `BBOX` or set it back to None.
method ... is not applicable	The method selects no IMERG cell for this basin: choose another one.
timestep(s) still missing	The download was interrupted: run the script again; files already downloaded are not downloaded twice.

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